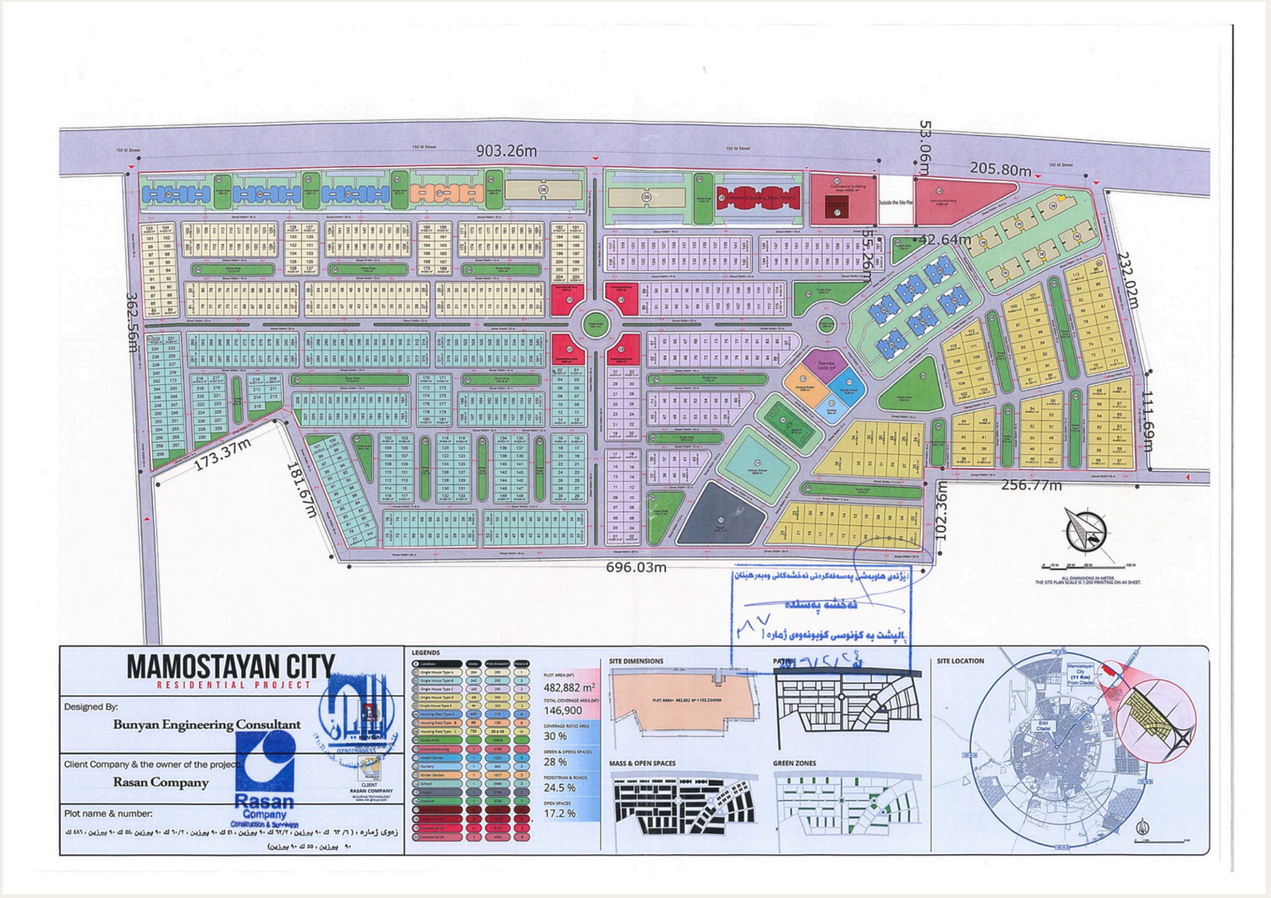
ARCHITECTURAL URBAN STUDY

ERBIL / KURDISTAN REGION / IRAQ

MAMOSTAYAN CITY



SMART URBAN GROWTH

Mixed Land Use

Compact Building Design

WALKABILITY

TRANSPORTATION

PREPARED BY

Location in Erbil



SHAWAYS

Mamostayan City is in the growing northern part of Erbil.

12.2 km

centre.

Main roads connect Mamostayan to other parts of Erbil.

Surroundings and regional connections



SITE EDGES

NORTH
150 m Road

NORTHEAST
Slava City

NORTHWEST
Spanish Village

EAST
Pirmam Road / Cavar City

SOUTH
Pirzin

Main roads make travel by car easier, but they can be difficult to cross on foot.

Project scale and land allocation

482,882 m²

146,900 m²
original plan.

30.4%
Share of the site covered by buildings. The plan rounds this to 30%.

PROJECT TEAM

Developer / owner
Rasan Company

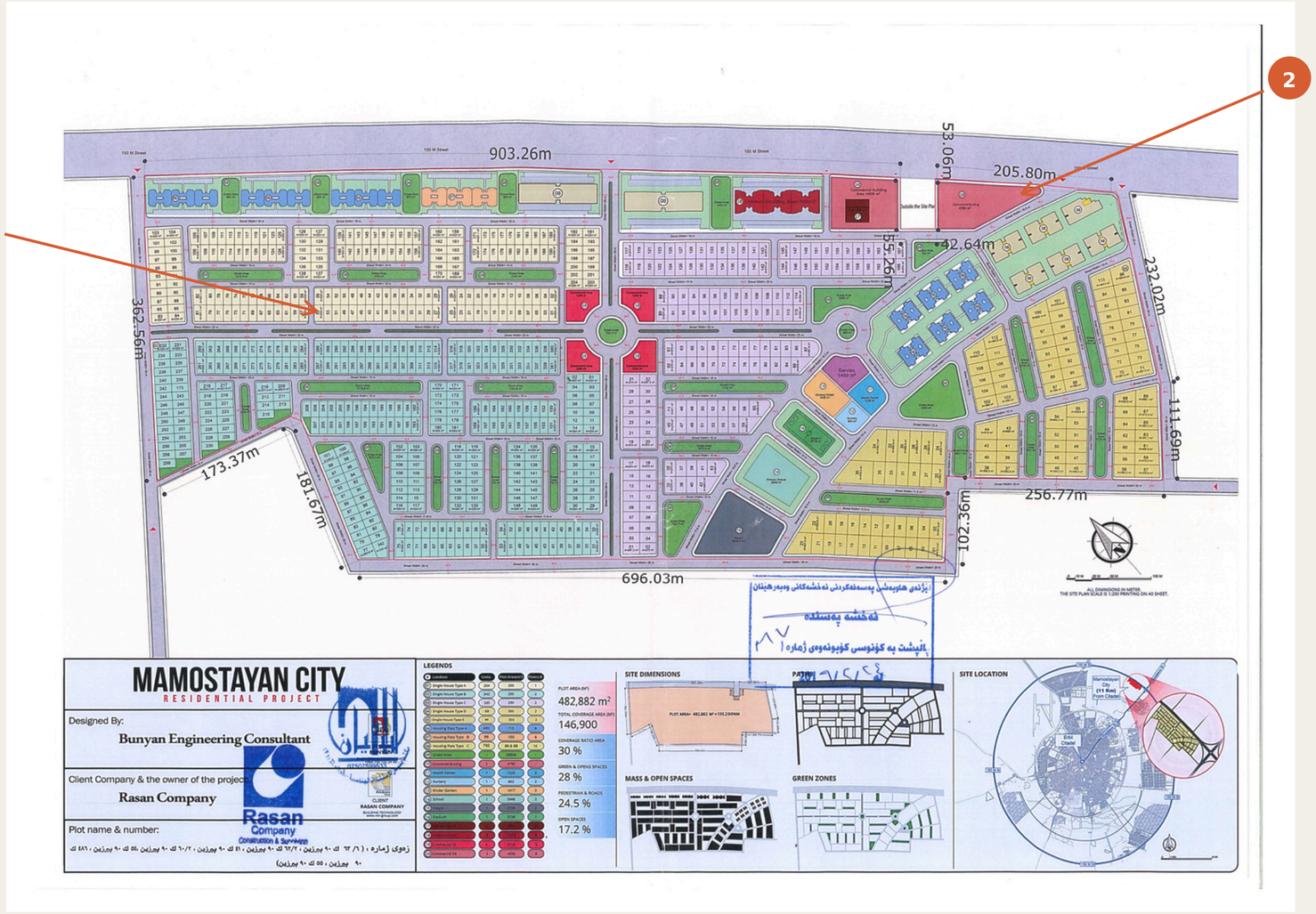
Master-plan designer
Bunyan Engineering Consultant

LAND ALLOCATION

28%	Green & open spaces
24.5%	Pedestrian & roads
17.2%	Open spaces

Buildings cover about one-third of the site. Roads, paths and open areas fill the spaces between the housing blocks.

Master plan and development changes



01 / ORIGINAL ORGANISATION

Most houses are in the inner blocks. Apartment buildings sit along parts of the outer edge. Community services are grouped in the south.

02 / UNBUILT COMMERCIAL LAND

Plot 10, the red triangle at the top right, is still empty because of problems with neighbouring owners.

03 / RASAN TOWERS

Rasan Towers add homes on land first planned for shops and businesses.

Housing choices and residential types



Housing type

LOW-RISE HOUSES

200 m² plots: 1 or 2 floors.
250 m² plots: 2 floors.



Housing type

LARGER VILLAS

350 m² plots: 2 floors.
Types D and E in the original plan.



Housing type

APARTMENT BUILDINGS

Types A and B: 8 floors.
Type C: 10 floors in the plan.

Rasan Towers add apartment choices of 75, 80, 110, 115, 135 and 150 m². These sizes give families more choice.

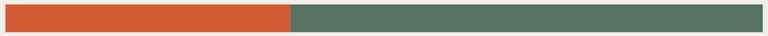
Planned housing capacity and density

ORIGINAL RESIDENTIAL SCHEDULE

TYPE	UNITS	AREA PER UNIT / PLOT	FLOORS
House A	204	200 m²	1
House B	342	200 m²	2
House C	165	250 m²	2
House D	69	350 m²	2
House E	44	350 m²	2
Flat A	480	115 m²	8
Flat B	96	150 m²	8
Flat C	**792**	**86 & 98 m²***	**10**

2,192

Total planned homes: 824 houses plus 1,368. apartments.



37.6% HOUSES 62.4% APARTMENTS

45.4

Planned homes per hectare for the entire site.

Apartments make up 62.4% of planned homes. The city offers both houses with private plots and apartments that use less land.

Green areas and usable open space



Open lawn between residential streets



Planted edge and paved path

28%

Land planned for green and open spaces.

SITE ANALYSIS

The shared lawn is close to homes. Plants and paths make it a pleasant place for short walks.

There are few seats, and some areas lack shade. More places to sit, rest and play would make the green spaces more useful.

Daily services and neighbourhood activity



Retail frontages

Shops provide daily needs close to home.



School

A school within the neighbourhood.



Kindergarten

A place for young children to learn.



Mosque

A place for prayer and community life.

These services support daily life. Safe walking routes are needed to connect them across the wide roads.

Streets, walking and parking



Pavement beside the kindergarten

WALKING

There are pavements, but the routes need more shade and safer crossings.



Parking between apartment buildings

SURFACE PARKING

Parking near entrances is useful, but large parking areas make walking less comfortable.



Ramp to basement parking

LAND EFFICIENCY

Parking below buildings saves space and leaves more land free at ground level.

PRINCIPLE 01

Mixed Land Use

Daily needs close to home

Shops and homes share buildings, with local services nearby. This mix helps people meet their daily needs close to home.

SCORING USED THROUGHOUT

1 = existing 0.5 = uncertain 0 = not existing
Each of the 20 policies gets one value.
The values show if a policy exists, not how good it is.



A Residential floors B Commercial base

Public support and development rules

MLU / POLICY 01

State support to live near work

Purpose. Use government funding to help people live closer to work.



Apartments and access to jobs

SITE ANALYSIS

The apartments offer more housing choices, but there is no government funding to help people live near their jobs. More homes alone do not reduce long trips to work.

NOT EXISTING

There is no funding programme to help people live near work.

0

MLU / POLICY 02

Codes that permit mixed use

Purpose. Allow homes and shops to share buildings and areas.



Shops below apartments

SITE ANALYSIS

Shops are on the ground floor, with apartments above. Smart Growth rules allow this mix. Residents can reach daily services close to home, and the shops keep the street busy.

EXISTING

Smart Growth rules allow homes and shops to be together.

1

Planning tools and investment

MLU / POLICY 03

Innovative planning tools

Purpose. Use planning tools to bring homes, shops and services together.



Shop area with housing behind

SITE ANALYSIS

The shop area and nearby apartments are planned together. Planning tools help bring these uses into one project. This makes local services easy for residents to reach.

EXISTING

Planning tools help bring homes and shops together. **1**

MLU / POLICY 04

Finance for mixed-use property

Purpose. Offer loans or financial help for buildings with more than one use.



Shops and homes in one building

SITE ANALYSIS

Some buildings have both shops and homes. However, there is no special loan or funding programme to help build more projects like these.

NOT EXISTING

Mixed-use buildings exist, but special financial help does not. **0**

Building form and flexibility

MLU / POLICY 05

Zoning by building form

Purpose. Control building size and shape while allowing different uses inside.



Shops following the same building line

SITE ANALYSIS

The shops follow the same building line and fit the size of nearby buildings. Different uses can work within this pattern, giving the street a clear and organised look.

EXISTING

The rules allow different uses within a clear building pattern.

1

MLU/POLICY 06

Flexible zoning

Purpose. Allow land and buildings to change use when needs change.



Towers within the housing area

SITE ANALYSIS

Flexible zoning allows changes in land use. Rasan Towers add homes on land first planned for shops and businesses. This shows how the plan can change as housing needs grow.

EXISTING

Land use can change to meet new needs.

1

Retail renewal and local scale

MLU / POLICY 07

Reuse declining retail

Purpose. Give old or unused shopping areas new uses.



Neighbourhood shops in use

SITE ANALYSIS

The shops are in use, and no declining shopping area needs to be changed. There is no project here that turns an old shopping site into a mix of new uses.

NOT EXISTING

No old shopping site has been changed to mixed use. **0**

MLU / POLICY 08

Mixed use at a suitable scale

Purpose. Keep shops and services at a size that suits the neighbourhood.



Small commercial units in a low-rise block

SITE ANALYSIS

The small shops fit well beside the houses and low-rise buildings. Their size suits daily needs. Residents have nearby services without large shops dominating the area.

EXISTING

The shops are a suitable size for the neighbourhood. **1**

Retrofitting and employment access

MLU / POLICY 09

Retrofit single-use development

Purpose. Improve existing commercial sites by adding uses and walking routes.



Commercial building with paving in front

SITE ANALYSIS

A wide paved area sits between the commercial building and the street. No existing commercial site has been changed into a walkable mixed-use place. The change at Rasan Towers is a change in land use.

NOT EXISTING

No existing commercial site has been improved in this way. **0**

MLU / POLICY 10

Reward a jobs-housing balance

Purpose. Reward neighbourhoods that bring jobs and homes closer together.



Housing beside local shops

SITE ANALYSIS

Some jobs are close to the homes, but most workplaces are farther away. Local shops provide a few nearby jobs. It is unclear if there is a reward for improving this balance.

UNCERTAIN

Some jobs are nearby, but any reward is still unclear. **0.5**

PRINCIPLE 02

Compact Building Design

More homes on less land, with shared outdoor space

Apartments fit more homes on less land by building upwards. Houses, roads and parking take up more ground space. These parts need to work well together.



More homes on the same piece of land

Using land well depends on buildings, streets and parking together.

Participation and open-space access

CBD / POLICY 01

Public education about density

Purpose. Hold meetings to discuss housing density with residents.



Community services within the city

SITE ANALYSIS

There are no public meetings about housing density. These meetings could help residents understand compact housing and discuss how it would affect daily life.

NOT EXISTING

There are no public meetings about compact housing. **0**

CBD / POLICY 02

Ready access to open space

Purpose. Provide shared outdoor spaces close to homes.



Open lawn directly beside housing

SITE ANALYSIS

Homes face a shared lawn with low plants around its edges. Residents can reach it easily from nearby streets. The green space is close enough for everyday use.

EXISTING

Shared open space is close to homes and easy to reach. **1**

Parking and the street section

CBD / POLICY 03

Parking with less land demand

Purpose. Save land by using basement, shared or street parking.



Parking ramp below the building

SITE ANALYSIS

Parking is available along streets and in separate parking areas. There is also parking below the apartments. This uses space under the buildings and reduces the land needed for cars outside.

EXISTING

Street, off-street and basement parking are available. **1**

CBD / POLICY 04

Building scale and street type

Purpose. Match building height and the space in front to the street.



Low-rise homes opposite apartment buildings

SITE ANALYSIS

The building heights and setbacks fit the streets. Taller apartment blocks have space in front, while houses keep a smaller scale. The space between buildings and roads gives good separation.

EXISTING

Building heights and setbacks suit the streets. **1**

Standards and density incentives

CBD / POLICY 05

Compact-design standards

Purpose. Use design rules that help fit more homes on less land.



Apartments with a similar building pattern

SITE ANALYSIS

Government design rules support compact housing. Apartment buildings fit several homes on the same piece of land. Their similar building lines also give the area an organised look.

EXISTING

Government rules support compact housing and good design.

1

CBD / POLICY 06

Density bonuses

Purpose. Allow extra floor space in return for benefits such as public space.



Higher residential buildings in the project

SITE ANALYSIS

The towers provide more homes by building upwards. However, there is no rule giving extra floor space in return for public benefits, such as better shared spaces.

NOT EXISTING

No extra floor space is offered in return for public benefits.

0

Privacy and design review

CBD / POLICY 07

Privacy in homes and yards

Purpose. Use walls, gates and space between buildings to protect privacy.



Boundary wall, gate and private entrance

SITE ANALYSIS

Walls, gates and plants separate private yards from the street. They reduce direct views into homes and give each house a private entrance.

EXISTING

Walls, gates and plants help keep homes and yards private.

1

CBD / POLICY 08

A design review board

Purpose. Have a board check designs before they are approved.



Houses with a similar design

SITE ANALYSIS

A design review board checks building design. Similar building fronts, materials and setbacks help the streets look organised. The buildings fit well together.

EXISTING

Design review helps buildings fit together.

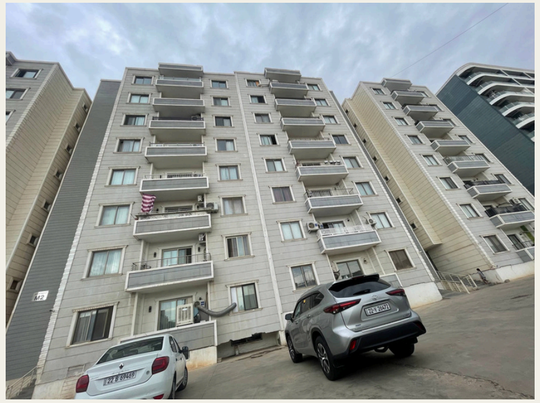
1

Financial support and regional planning

CBD / POLICY 09

Financial incentives for density

Purpose. Offer financial help for housing that uses less land.



Apartments on the same piece of land

SITE ANALYSIS

The apartment blocks fit several homes on one piece of land. However, there are no special grants or lower fees to encourage this type of housing.

NOT EXISTING

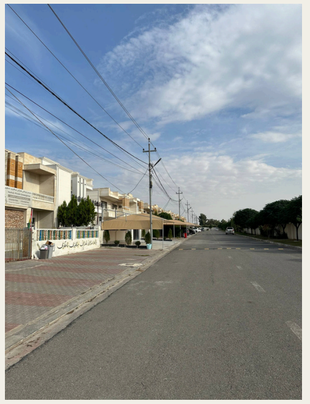
There is no special financial help for higher-density housing.

0

CBD / POLICY 10

Community support for regional plans

Purpose. Involve residents in plans for the wider city.



Street at the edge of the development

SITE ANALYSIS

The streets connect Mamostayan to the wider city. Planning these areas together is important, but residents have no organised role in the wider planning process.

NOT EXISTING

Residents have no organised role in planning the wider area.

0

Final assessment: policy status

POLICY	01	02	03	04	05	06	07	08	09	10	
Mixed Land Use	0	1	1	0	1	1	0	1	0	0.5	**5.5 points**
Compact Design	0	1	1	1	1	0	1	1	0	0	**6 points**

11.5 points
11 existing + 1 uncertain
+ 8 not existing policies

Existing 11 of 20 policies = 55%
Uncertain 1 of 20 policies = 5%
Not existing 8 of 20 policies = 40%

Total = (11 × 1) + (1 × 0.5) + (8 × 0) = 11.5.
11 policies exist, 1 is unclear and 8 do not exist. This total shows their status. It is not a percentage score.

MLU 07 gets 0 because no old shopping site has been reused. The shops are in use, so there is no clear need for this type of project.

Design priorities for the next stage



Walking route beside apartments

Priority: improve walking routes as buildings get taller.

01 / CONTINUOUS WALKING ROUTES

Connect homes, shops and schools with clear paths, shade and raised crossings. Keep ramps and parked cars out of walking routes.

02 / MORE USEFUL SHARED SPACE

Add seats and shade to the lawns. Connect green spaces to play areas, shops and services so people can use them every day.

03 / COMPACTNESS WITH PUBLIC BENEFIT

Check parking needs before paving more land. When adding more homes, also improve shared spaces. Include residents in decisions about their neighbourhood.

Conclusion
Homes, shops and open spaces that support daily life.

Mamostayan brings homes, shops and open spaces together. Nearby services, different housing choices and privacy are its main strengths. More shade, safer paths and a bigger role for residents would make daily life better.

EXISTING POLICIES

10 / 20

WHAT WORKS WELL

01 **Mixed-use streets**
Shops and homes share buildings, keeping daily needs close to residents.

02 **Compact housing**
Apartments fit more homes on the same piece of land.

03 **Open space and privacy**
Residents have shared lawns as well as private entrances and yards.

04 **A clear next step**
Make walking easier and involve residents in future plans.

Walkability

Make daily trips easy on foot

Nearby shops, schools and green spaces give people reasons to walk. The next step is to make the routes between them safer, clearer and easier for everyone to use.



A direct walking path towards the mosque

Funding and nearby services

WALK / POLICY 01

Funding for better walking routes

Purpose. Give communities grants to improve existing streets and pavements.



Damaged edge beside the apartments

SITE ANALYSIS

The pavement beside the apartments has damaged edges and loose materials, making walking less safe. Repairs are needed, but government funding for these improvements is not confirmed.

UNSURE

Sidewalk improvements are needed, but financial support is unclear.

0.5

WALK/POLICY 02

Daily services close to homes

Purpose. Group key services near homes, jobs and public transport.



Kindergarten beside apartment buildings

SITE ANALYSIS

The kindergarten sits close to apartment blocks. Shops and other services are also within the neighbourhood. This supports short local trips,

AVAILABLE

Services are near homes.

1

Building fronts and safe streets

WALK / POLICY 03

Buildings that support walking

Purpose. Require shop fronts and entrances that make walking easy.



Shop fronts facing the street

SITE ANALYSIS

Ground-floor shops give people places to walk to. However, parked cars and changes in level make some entrances harder to reach.

UNSURE

Active shop fronts exist, but easy access is not consistent.

0.5

WALK / POLICY 04

Safe street design standards

Purpose. Set street rules that protect people walking and cycling.



Street and pavements beside the kindergarten

SITE ANALYSIS

The street has sidewalks, but lacks important safety features such as pedestrian crossings, bicycle lanes, and visible traffic-calming measures. This creates safety concerns for pedestrians, especially near the kindergarten.

NOT EXSTING

Proper pedestrian and cycling safety measures are not provided.

0

Sidewalk standards and slower traffic

WALK / POLICY 05

Sidewalk design standards

Purpose. Set rules for pavement width, surfaces, ramps and clear paths.



Paved path along the kindergarten boundary

SITE ANALYSIS

The pavements gives a clear route beside the kindergarten, and other places. Some areas have rough edges or obstacles.
.

UNSURE

Pavements with standard exist **1**

WALK / POLICY 06

Traffic calming

Purpose. Use speed humps and other measures where cars travel too fast.



Speed hump across a residential street

SITE ANALYSIS

A speed hump crosses the residential street. It makes drivers slow down near homes and supports safer local movement. This is a clear traffic-calming measure within the neighbourhood.

AVAILABLE

speed humps are available to slow traffic near homes. **1**

Better paths and access for everyone

WALK / POLICY 07

Attractive and cared-for paths

Purpose. Plant, improve and maintain walking routes.



Paved path through a planted area

SITE ANALYSIS

Trees, grass and trimmed shrubs line this paved path. The planting makes the route more pleasant and gives some shade. Similar care would help the rougher walking routes elsewhere.

AVAILABLE

Planted and cared-for walking paths are available. **1**

WALK / POLICY 08

Access for people with disabilities

Purpose. Provide step-free access to paths, streets, parks and services.



Steps and obstacles in front of shops

SITE ANALYSIS

The shop entrances have steps, and items narrow the space in front. A clear, step-free route is not available at these entrances. Ramps and an open path are needed for wheelchair access.

NOT AVAILABLE

Step-free access is not available at the reviewed shop entrances. **0**

Walking links and local activity

WALK / POLICY 09

Connect walking routes

Purpose. Join paths, parking areas, green routes and buildings.



Green strip between shops and parking

SITE ANALYSIS

The green strip sits between shops, homes and parking. These uses are close together, but curbs and road crossings interrupt the walking route. A fully connected path network remains unclear.

UNSURE

Places are close together. Continuous walking links arent existed.

0.5

WALK / POLICY 10

Local business and street activity

Purpose. Support shops and activities that give people reasons to walk.



Local shops and outdoor seating

SITE ANALYSIS

Small shops and outdoor seating create places to visit near home. They bring everyday activity to the street and give residents reasons to make short trips on foot.

AVAILABLE

Local businesses create destinations and activity for walking.

1

Transportation Options

More ways to reach everyday places

HOW THE NEW POLICIES ARE RATED

1 = available 0.5 = unsure 0 = not available
An unclear policy receives 0.5, even when the street or building itself is visible.



Cycling and car access beside neighbourhood shops

Transport funding and street standards

TRANS / POLICY 01

Funding for more travel choices

Purpose. Fund walking, cycling and public transport with suitable land use.



Homes, services and streets close together

SITE ANALYSIS

The road mainly supports car movement. No dedicated cycling lanes or public transport facilities are visible. Although residential and commercial buildings are nearby, improvements supporting alternative transportation are not clearly provided.

NOT EXISTING

No visible improvements supporting different transportation choices.

0

TRANS / POLICY 02

Street standards near transit

Purpose. Adjust traffic-flow standards where public transport is available.



Road leading towards the apartment area

SITE ANALYSIS

The road is mainly designed for cars. No bus stops, dedicated transit facilities, or special street features for public transport are visible. The street does not show adjustments to support public transportation.

NOT EXISTING

No visible implementation of street standards near public transport.

0

Street networks and travel connections

TRANS / POLICY 03

Connected local streets

Purpose. Plan connected two- or four-lane streets with short blocks.



Junction between residential streets

SITE ANALYSIS

Local streets meet at regular junctions. The master plan uses small housing blocks and several links between them. This gives residents a choice of routes through the neighbourhood.

AVAILABLE

Connected local streets and small housing blocks are available.

1

TRANS / POLICY 04

Connect different ways to travel

Purpose. Make it easy to change between walking, cycling and transport.



People, cars and a motorcycle near shops

SITE ANALYSIS

The commercial area supports cars, motorcycles, and pedestrians, but there are no visible dedicated cycling lanes, public transport stops, or organized connections between different transportation modes. Movement mainly depends on private vehicles.

NOT EXISTING

Several travel modes are present. Their organised links are unsure.

0

Transit centres and new sidewalks

TRANS / POLICY 05

Activity centres around transit

Purpose. Use zoning to group busy places around public transport stops.



Shops and apartments around a central space

SITE ANALYSIS

The area contains shops and apartments that form a local activity centre. However, the centre is not organized around public transport stops, and access mainly depends on private vehicles.

NOT EXISTING

The activity centre is not connected to a planned public transportation system.

0

TRANS / POLICY 06

Sidewalks in new development

Purpose. Require pavements in every new development.



Pavement beside the school boundary

SITE ANALYSIS

All residential streets have paved sidewalks, providing safe walking spaces for pedestrians. Sidewalks are consistently included throughout the development.

EXISTING

Sidewalks are provided throughout the neighborhood.

1

Parking and workplace travel

TRANS / POLICY 07

Plan for parking needs

Purpose. Provide suitable parking and use land carefully.



Basement entrance and parking beside shops

SITE ANALYSIS

The development provides street, off-street and basement parking. The basement uses space below the building and reduces the land needed for cars outside. Walking routes around parking still need care.

AVAILABLE

Several parking options are available within the development.

1

TRANS/POLICY 08

Workplace travel programmes

Purpose. Work with employers to reduce car trips during busy hours.



Parking serving the apartment area

SITE ANALYSIS

Many private cars are parked near the apartments, showing car use in the area. However, there is no confirmed information about employer-provided buses, shared transport, or flexible working hours.

UNSURE

Workplace travel programmes have not been confirmed.

0.5

Transit services and regional freight

TRANS / POLICY 09

Transit that serves the neighbourhood

Purpose. Adjust public transport routes and services to suit local needs.



Apartment and shop frontage along a road

SITE ANALYSIS

Shari Mamostayan has apartments and shops, but there are no organized public transport routes or bus stops serving the neighborhood. Residents mainly depend on private cars and taxis for transportation.

NOT EXISTING

No public transport services are provided to meet neighborhood needs.

0

TRANS / POLICY 10

Group freight near major terminals

Purpose. Place freight sites near ports, airports or rail terminals.



Residential streets at the city edge

SITE ANALYSIS

Shari Mamostayan is mainly a residential area with no warehouses or freight facilities. Therefore, this policy is not directly applicable to the neighborhood.

NOT EXISTING

No freight facilities exist within the study area.

0

40 POLICIES / CORRECTED RATING SCALE

Final assessment: four principles

POLICY	01	02	03	04	05	06	07	08	09	10	SCORE / 10
Mixed Land Use	0	1	1	0	1	1	0	1	0	0.5	5.5
Compact Design	0	1	1	1	1	0	1	1	0	0	6
Walkability	0.5	1	0.5	0	1	1	1	0	0.5	1	6.5
Transportation	0	0	1	0	0	1	1	0.5	0	0	3.5

21.5 /40

TOTAL POLICY SCORE

19 available + 5 unsure + 16 not available

RATING KEY

1 Available

0.5 Unsure

0 Not available

Half points mark unsure policies. They do not confirm that a policy is available.

Design priorities for the next stage



Green space beside a residential street

Make the short trip safe, shaded and easy for everyone.

01 / COMPLETE THE WALKING ROUTES

Repair broken paths, keep them clear and add ramps at steps and curbs. Use safe crossings to connect homes, shops, schools and green spaces.

02 / MAKE DAILY WALKING MORE COMFORTABLE

Add shade and seating. Keep parked cars away from paths and shop entrances. Extend the care seen in planted areas to the rougher routes.

03 / LINK THE NEIGHBOURHOOD TO WIDER TRANSPORT

Confirm public transport routes and stops, then connect them to local paths. Explore safe cycle links and employer travel programmes with residents.

Conclusion
The places are close. The connections can be better.

Mamostayan combines homes, shops and open spaces. Connected streets and parking support daily life. Better walking access and clearer transport links would help these places work together.

TOTAL POLICY SCORE

21.5 / 40

FOUR PRINCIPLES / ONE NEIGHBOURHOOD

01 **Mixed Land Use**
Homes and local shops share space and bring daily needs closer.

02 **Compact Building Design**
Apartments use land efficiently, with shared lawns and private homes nearby.

03 **Walkability**
Paths and speed humps help. Steps, gaps and obstacles still need attention.

04 **Transportation Options**
Local streets and parking are clear strengths. Wider travel choices remain unsure.